

Отчёт по лабораторной работе №6

Адресация IPv4 и IPv6. Двойной стек

Чилеше Лупупа

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Российский университет дружбы народов, Москва, Россия

Цель работы

Изучение принципов распределения и настройки адресного пространства в IPv4/IPv6 сетях, а также проверка работы сети с двойным стеком адресации.

Разбиение IPv4-сети на подсети

Сеть 172.16.20.0/24 (характеристики)

- Префикс: /24, маска: 255.255.255.0
- Сеть: 172.16.20.0, broadcast: 172.16.20.255
- Хосты: 172.16.20.1 – 172.16.20.254 (254 узлов)

Разбиение 172.16.20.0/24 на 126, 62, 62 (VLSM)

- 126 узлов → /25
- 62 узла → /26
- 62 узла → /26

Подсеть	Префикс	Сеть	Broadcast	Диапазон хостов
1	/25	172.16.20.0	172.16.20.127	172.16.20.1 – 172.16.20.126
2	/26	172.16.20.128	172.16.20.191	172.16.20.129 – 172.16.20.190
3	/26	172.16.20.192	172.16.20.255	172.16.20.193 – 172.16.20.254

Исходная сеть: - /26, маска 255.255.255.192 - сеть 10.10.1.64, broadcast 10.10.1.127 - хосты 10.10.1.65 – 10.10.1.126 (62 узла)

Подсеть на 30 узлов: - /27, сеть 10.10.1.64/27 - broadcast 10.10.1.95 - хосты 10.10.1.65 – 10.10.1.94

Исходная сеть: - /26, маска 255.255.255.192 - сеть 10.10.1.0, broadcast 10.10.1.63 - хосты 10.10.1.1 – 10.10.1.62 (62 узла)

Подсеть на 14 узлов: - /28, сеть 10.10.1.0/28 - broadcast 10.10.1.15 - хосты 10.10.1.1 – 10.10.1.14

Разбиение IPv6-сети на подсети

- Префикс: /48
- Маска: ffff:ffff:ffff::
- Диапазон (краевые значения):
 - min: 2001:db8:c0de:0000:0000:0000:0000:0000
 - max: 2001:db8:c0de:ffff:ffff:ffff:ffff:ffff

Разбиение по Subnet ID (добавляем 1 бит к префиксу):

- Подсеть A: 2001:db8:c0de:0000::/49
- Подсеть B: 2001:db8:c0de:8000::/49

Смысл: деление выполняется в области Subnet ID (четвёртый хекстет), сохраняя иерархию /48 как site prefix.

Разбиение внутри host-части:

- Подсеть A: 2001:db8:c0de:0000::/65
- Подсеть B: 2001:db8:c0de:0000:8000::/65

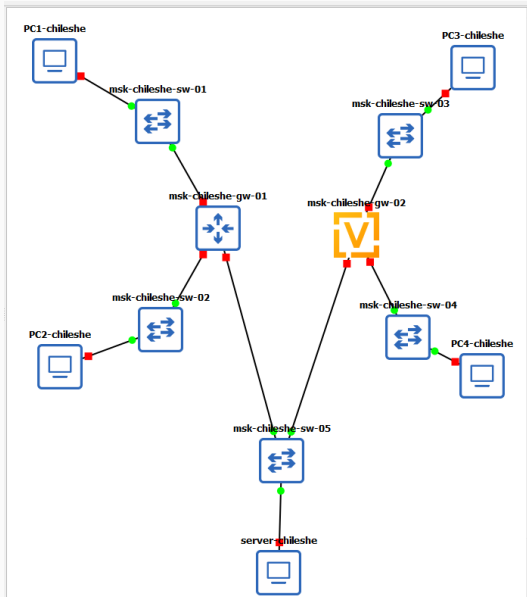
Смысл: деление происходит в Interface ID (последние 64 бита). Практически используется редко из-за ожидания /64 (SLAAC).

- Префикс: /64
- Маска: ffff:ffff:ffff:ffff::
- Диапазон (краевые значения):
 - min: 2a02:6b8:0000:0000:0000:0000:0000:0000
 - max: 2a02:6b8:0000:0000:ffff:ffff:ffff:ffff

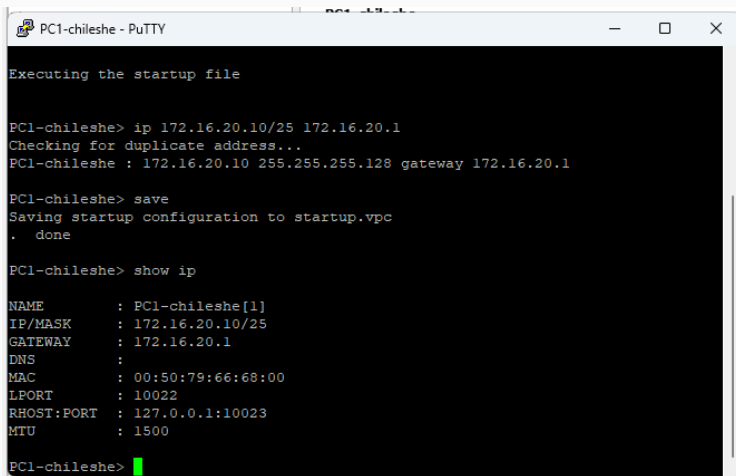
Разбиение на 2 подсети возможно только с изменением префикса (например /65), то есть фактически — по Interface ID: - 2a02:6b8::/65 - 2a02:6b8:0:0:8000::/65

Адресация IPv4 и IPv6. Двойной стек

Топология лабораторной работы в GNS3



Настройка IPv4 на PC1



```
PC1-chileshe - PuTTY

Executing the startup file

PC1-chileshe> ip 172.16.20.10/25 172.16.20.1
Checking for duplicate address...
PC1-chileshe : 172.16.20.10 255.255.255.128 gateway 172.16.20.1

PC1-chileshe> save
Saving startup configuration to startup.vpc
. done

PC1-chileshe> show ip

NAME       : PC1-chileshe[1]
IP/MASK    : 172.16.20.10/25
GATEWAY    : 172.16.20.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10022
RHOST:PORT : 127.0.0.1:10023
MTU        : 1500

PC1-chileshe> 
```

Рис. 2: Настройка IPv4-адреса на PC1 и show ip

Настройка IPv4 на PC2



```
PC2-chileshe - PuTTY

Executing the startup file

PC2-chileshe> ip 172.16.20.138/25 172.16.20.129
Checking for duplicate address...
PC2-chileshe : 172.16.20.138 255.255.255.128 gateway 172.16.20.129

PC2-chileshe> save
Saving startup configuration to startup.vpc
. done

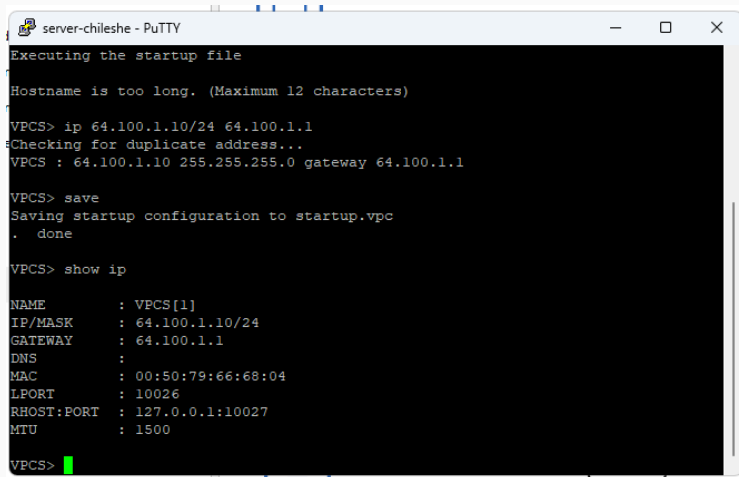
PC2-chileshe> show ip

NAME       : PC2-chileshe[1]
IP/MASK    : 172.16.20.138/25
GATEWAY    : 172.16.20.129
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 10024
RHOST:PORT : 127.0.0.1:10025
MTU        : 1500

PC2-chileshe>
```

Рис. 3: Настройка IPv4-адреса на PC2 и show ip

Настройка IPv4 на сервере (Dual Stack Server)



```
server-chileshe - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 64.100.1.10/24 64.100.1.1
Checking for duplicate address...
VPCS : 64.100.1.10 255.255.255.0 gateway 64.100.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

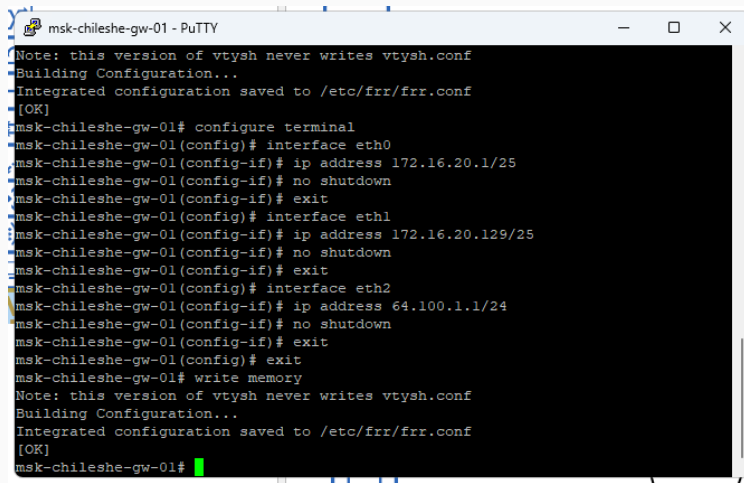
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 64.100.1.10/24
GATEWAY    : 64.100.1.1
DNS        :
MAC        : 00:50:79:66:68:04
LPORT      : 10026
RHOST:PORT : 127.0.0.1:10027
MTU        : 1500

VPCS> 
```

Рис. 4: Настройка IPv4 на сервере и show ip

Настройка IPv4 на маршрутизаторе FRR



```
msk-chileshe-gw-01 - PuTTY
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-chileshe-gw-01# configure terminal
msk-chileshe-gw-01(config)# interface eth0
msk-chileshe-gw-01(config-if)# ip address 172.16.20.1/25
msk-chileshe-gw-01(config-if)# no shutdown
msk-chileshe-gw-01(config-if)# exit
msk-chileshe-gw-01(config)# interface eth1
msk-chileshe-gw-01(config-if)# ip address 172.16.20.129/25
msk-chileshe-gw-01(config-if)# no shutdown
msk-chileshe-gw-01(config-if)# exit
msk-chileshe-gw-01(config)# interface eth2
msk-chileshe-gw-01(config-if)# ip address 64.100.1.1/24
msk-chileshe-gw-01(config-if)# no shutdown
msk-chileshe-gw-01(config-if)# exit
msk-chileshe-gw-01(config)# exit
msk-chileshe-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-chileshe-gw-01#
```

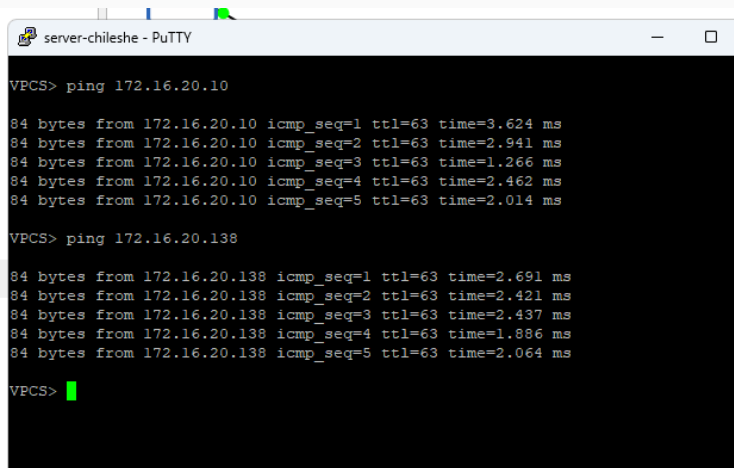
Рис. 5: Настройка интерфейсов FRR (eth0/eth1/eth2)

Проверка конфигурации FRR

```
msk-chileshe-gw-01 - PuTTY
[OK]
msk-chileshe-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.1
frr defaults traditional
hostname frr
hostname msk-chileshe-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 172.16.20.1/25
exit
!
interface eth1
 ip address 172.16.20.129/25
exit
!
interface eth2
 ip address 64.100.1.1/24
exit
!
end
msk-chileshe-gw-01# show interface brief
Interface      Status   VRF      Addresses
-----
eth0            up       default  172.16.20.1/25
eth1            up       default  172.16.20.129/25
eth2            up       default  64.100.1.1/24
eth3            down     default
eth4            down     default
eth5            down     default
eth6            down     default
eth7            down     default
lo              up       default
pimreg         up       default
msk-chileshe-gw-01#
```

Проверка связности IPv4 (ping с сервера)



```
server-chileshe - PuTTY

VPCS> ping 172.16.20.10

84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=3.624 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=2.941 ms
84 bytes from 172.16.20.10 icmp_seq=3 ttl=63 time=1.266 ms
84 bytes from 172.16.20.10 icmp_seq=4 ttl=63 time=2.462 ms
84 bytes from 172.16.20.10 icmp_seq=5 ttl=63 time=2.014 ms

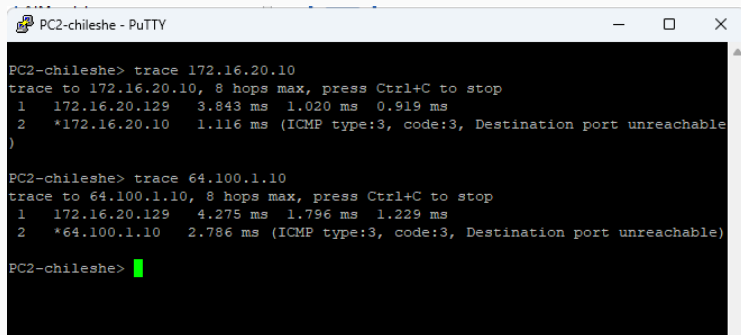
VPCS> ping 172.16.20.138

84 bytes from 172.16.20.138 icmp_seq=1 ttl=63 time=2.691 ms
84 bytes from 172.16.20.138 icmp_seq=2 ttl=63 time=2.421 ms
84 bytes from 172.16.20.138 icmp_seq=3 ttl=63 time=2.437 ms
84 bytes from 172.16.20.138 icmp_seq=4 ttl=63 time=1.886 ms
84 bytes from 172.16.20.138 icmp_seq=5 ttl=63 time=2.064 ms

VPCS> █
```

Рис. 7: Ping до PC1 и PC2 с сервера

Проверка маршрута IPv4 (trace с PC2)



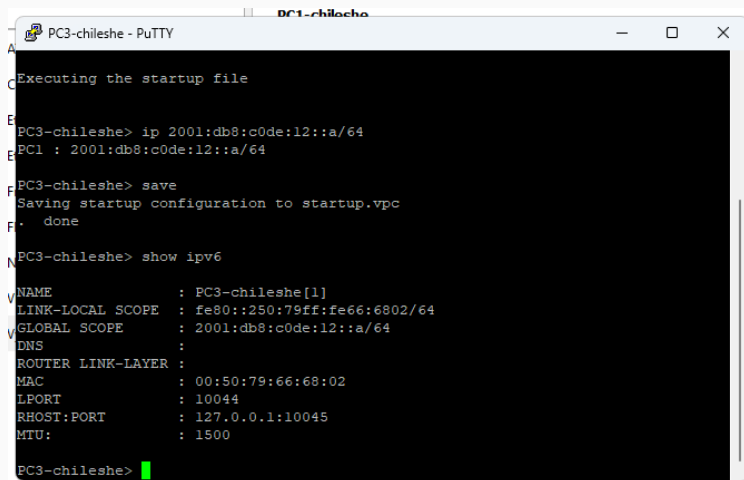
```
PC2-chileshe> trace 172.16.20.10
trace to 172.16.20.10, 8 hops max, press Ctrl+C to stop
 1  172.16.20.129    3.843 ms  1.020 ms  0.919 ms
 2  *172.16.20.10   1.116 ms (ICMP type:3, code:3, Destination port unreachable)
)

PC2-chileshe> trace 64.100.1.10
trace to 64.100.1.10, 8 hops max, press Ctrl+C to stop
 1  172.16.20.129    4.275 ms  1.796 ms  1.229 ms
 2  *64.100.1.10    2.786 ms (ICMP type:3, code:3, Destination port unreachable)

PC2-chileshe> █
```

Рис. 8: Trace с PC2 до PC1 и сервера

Настройка IPv6 на PC3



```
PC3-chileshe - PuTTY
Executing the startup file

PC3-chileshe> ip 2001:db8:c0de:12::a/64
PC1 : 2001:db8:c0de:12::a/64

PC3-chileshe> save
Saving startup configuration to startup.vpc
. done

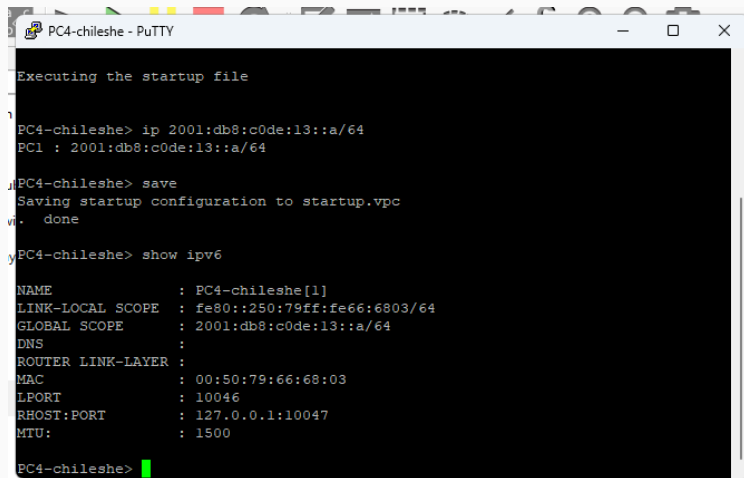
PC3-chileshe> show ipv6

NAME                : PC3-chileshe[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6802/64
GLOBAL SCOPE        : 2001:db8:c0de:12::a/64
DNS                  :
ROUTER LINK-LAYER   :
MAC                  : 00:50:79:66:68:02
LPORT                : 10044
RHOST:PORT           : 127.0.0.1:10045
MTU                  : 1500

PC3-chileshe>
```

Рис. 9: Настройка IPv6 на PC3 и show ipv6

Настройка IPv6 на PC4



```
PC4-chileshe - PuTTY

Executing the startup file

PC4-chileshe> ip 2001:db8:c0de:13::a/64
PC1 : 2001:db8:c0de:13::a/64

PC4-chileshe> save
Saving startup configuration to startup.vpc
vi. done

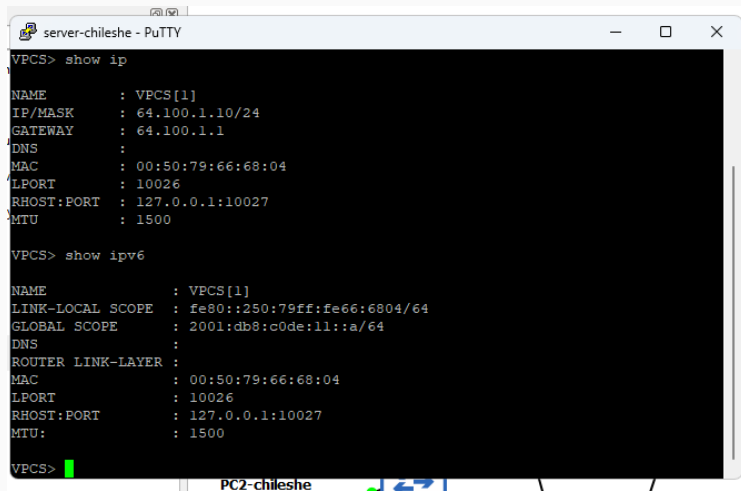
PC4-chileshe> show ipv6

NAME                : PC4-chileshe[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6803/64
GLOBAL SCOPE        : 2001:db8:c0de:13::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:03
LPORT               : 10046
RHOST:PORT           : 127.0.0.1:10047
MTU                 : 1500

PC4-chileshe>
```

Рис. 10: Настройка IPv6 на PC4 и show ipv6

Проверка IPv4/IPv6 на сервере



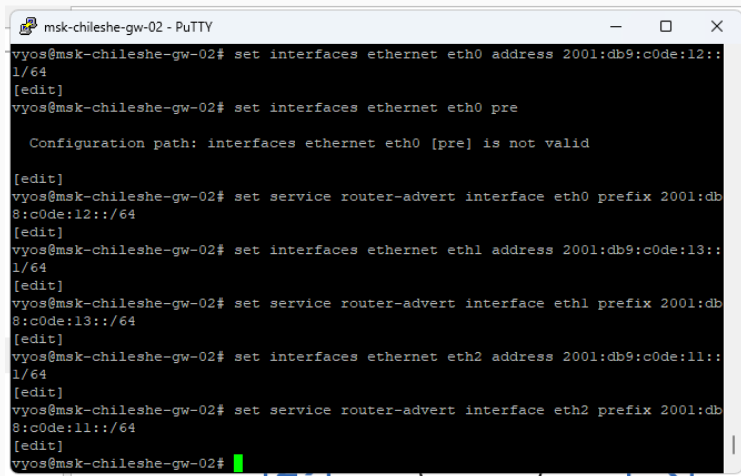
```
server-chileshe - PuTTY
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 64.100.1.10/24
GATEWAY    : 64.100.1.1
DNS        :
MAC        : 00:50:79:66:68:04
LPORT     : 10026
RHOST:PORT : 127.0.0.1:10027
MTU        : 1500

VPCS> show ipv6
NAME      : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6804/64
GLOBAL SCOPE    : 2001:db8:c0de:11::a/64
DNS             :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:04
LPORT         : 10026
RHOST:PORT     : 127.0.0.1:10027
MTU            : 1500

VPCS>
```

Рис. 11: show ip и show ipv6 на сервере двойного стека

Настройка IPv6 на VyOS (msk-chileshe-gw-02)



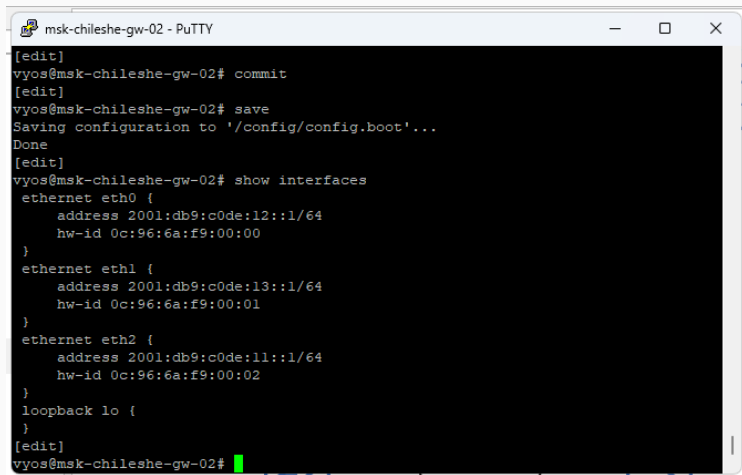
```
msk-chileshe-gw-02 - PuTTY
vyos@msk-chileshe-gw-02# set interfaces ethernet eth0 address 2001:db9:c0de:12::
1/64
[edit]
vyos@msk-chileshe-gw-02# set interfaces ethernet eth0 pre

Configuration path: interfaces ethernet eth0 [pre] is not valid

[edit]
vyos@msk-chileshe-gw-02# set service router-advert interface eth0 prefix 2001:db
8:c0de:12::/64
[edit]
vyos@msk-chileshe-gw-02# set interfaces ethernet eth1 address 2001:db9:c0de:13::
1/64
[edit]
vyos@msk-chileshe-gw-02# set service router-advert interface eth1 prefix 2001:db
8:c0de:13::/64
[edit]
vyos@msk-chileshe-gw-02# set interfaces ethernet eth2 address 2001:db9:c0de:11::
1/64
[edit]
vyos@msk-chileshe-gw-02# set service router-advert interface eth2 prefix 2001:db
8:c0de:11::/64
[edit]
vyos@msk-chileshe-gw-02#
```

Рис. 12: Назначение IPv6 адресов и Router Advertisement на VyOS

Проверка интерфейсов VyOS



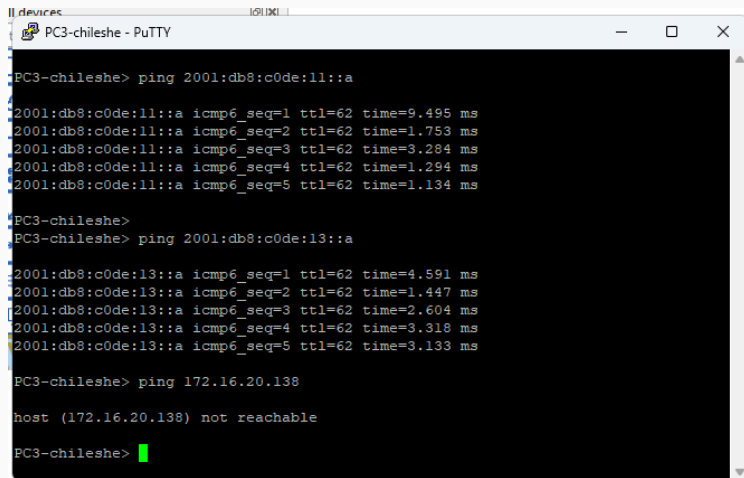
```
msk-chileshe-gw-02 - PuTTY
[edit]
vyos@msk-chileshe-gw-02# commit
[edit]
vyos@msk-chileshe-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-02# show interfaces
  ethernet eth0 {
    address 2001:db9:c0de:12::1/64
    hw-id 0c:96:6a:f9:00:00
  }
  ethernet eth1 {
    address 2001:db9:c0de:13::1/64
    hw-id 0c:96:6a:f9:00:01
  }
  ethernet eth2 {
    address 2001:db9:c0de:11::1/64
    hw-id 0c:96:6a:f9:00:02
  }
  loopback lo {
  }
[edit]
vyos@msk-chileshe-gw-02#
```

Рис. 13: show interfaces на VyOS

```
[edit]
vyos@msk-chileshe-gw-02# show interfaces
  ethernet eth0 {
    address 2001:db8:c0de:12::1/64
    hw-id 0c:96:6a:f9:00:00
  }
  ethernet eth1 {
    address 2001:db8:c0de:13::1/64
    hw-id 0c:96:6a:f9:00:01
  }
  ethernet eth2 {
    address 2001:db8:c0de:11::1/64
    hw-id 0c:96:6a:f9:00:02
  }
  loopback lo {
  }
[edit]
vyos@msk-chileshe-gw-02#
```

Рис. 14: IPv6 ping успешен, IPv4 из IPv6-сети недоступен

Сервер двойного стека доступен в обе стороны



```
PC3-chileshe> ping 2001:db8:c0de:11::a

2001:db8:c0de:11::a icmp6_seq=1 ttl=62 time=9.495 ms
2001:db8:c0de:11::a icmp6_seq=2 ttl=62 time=1.753 ms
2001:db8:c0de:11::a icmp6_seq=3 ttl=62 time=3.284 ms
2001:db8:c0de:11::a icmp6_seq=4 ttl=62 time=1.294 ms
2001:db8:c0de:11::a icmp6_seq=5 ttl=62 time=1.134 ms

PC3-chileshe>
PC3-chileshe> ping 2001:db8:c0de:13::a

2001:db8:c0de:13::a icmp6_seq=1 ttl=62 time=4.591 ms
2001:db8:c0de:13::a icmp6_seq=2 ttl=62 time=1.447 ms
2001:db8:c0de:13::a icmp6_seq=3 ttl=62 time=2.604 ms
2001:db8:c0de:13::a icmp6_seq=4 ttl=62 time=3.318 ms
2001:db8:c0de:13::a icmp6_seq=5 ttl=62 time=3.133 ms

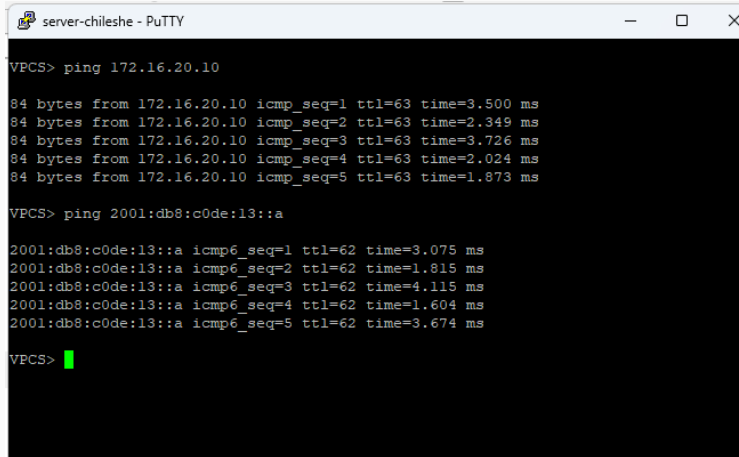
PC3-chileshe> ping 172.16.20.138

host (172.16.20.138) not reachable

PC3-chileshe> █
```

Рис. 15: Ping по IPv4 и IPv6 с сервера

Изоляция IPv4 и IPv6 подсетей



```
server-chileshe - PuTTY

VPCS> ping 172.16.20.10

84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=3.500 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=2.349 ms
84 bytes from 172.16.20.10 icmp_seq=3 ttl=63 time=3.726 ms
84 bytes from 172.16.20.10 icmp_seq=4 ttl=63 time=2.024 ms
84 bytes from 172.16.20.10 icmp_seq=5 ttl=63 time=1.873 ms

VPCS> ping 2001:db8:c0de:13::a

2001:db8:c0de:13::a icmp6_seq=1 ttl=62 time=3.075 ms
2001:db8:c0de:13::a icmp6_seq=2 ttl=62 time=1.815 ms
2001:db8:c0de:13::a icmp6_seq=3 ttl=62 time=4.115 ms
2001:db8:c0de:13::a icmp6_seq=4 ttl=62 time=1.604 ms
2001:db8:c0de:13::a icmp6_seq=5 ttl=62 time=3.674 ms

VPCS> █
```

Рис. 16: Подтверждение изоляции подсетей

Анализ трафика (Wireshark)

ARP: разрешение MAC по IPv4

```
15 9.6/0285 2001:db8:c0de:11::8 2001:db8:c0de:12::8 ICMPv6 118 Echo (ping) reply id=0xb10, seq=5, hop limit=65 (request in 12)
14 57.241875 Private_66:68:04 Broadcast ARP 64 Who has 64.100.1.1? Tell 64.100.1.10
15 57.246467 0c:66:71:16:00:02 Private_66:68:04 ARP 60 64.100.1.1 is at 0c:66:71:16:00:02
16 57.248178 64.100.1.10 172.16.20.10 ICMP 98 Echo (ping) request id=0xb10, seq=1/256, ttl=64 (reply in 17)

Frame 14: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
  Ethernet II, Src: Private_66:68:04 (00:50:79:66:68:04), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
    Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    Source: Private_66:68:04 (00:50:79:66:68:04)
      Type: ARP (0x0806)
      [Stream index: 3]
      Padding: 00000000000000000000000000000000
      Frame check sequence: 0x00000000 [unverified]
      [FCS Status: Unverified]
  Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Private_66:68:04 (00:50:79:66:68:04)
    Sender IP address: 64.100.1.10
    Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
    Target IP address: 64.100.1.1
```

Рис. 17: ARP-запрос к шлюзу 64.100.1.1

ICMP: Echo request / reply (IPv4)

No.	Time	Source	Destination	Protocol	Length	Info		
14	57.241875	Private_66:68:04	Broadcast	ARP	64	Who has 64.100.1.1? Tell 64.100.1.10		
15	57.246467	0c:66:71:16:00:02	Private_66:68:04	ARP	60	64.100.1.1 is at 0c:66:71:16:00:02		
---	16	57.248178	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request	id=0xbe10, seq=1/256, ttl=64 (reply in 17)
---	17	57.251254	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply	id=0xbe10, seq=1/256, ttl=63 (request in 16)
18	58.253149	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request	id=0xbf10, seq=2/512, ttl=64 (reply in 19)	
19	58.254944	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply	id=0xbf10, seq=2/512, ttl=63 (request in 18)	
20	59.255779	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request	id=0xc010, seq=3/768, ttl=64 (reply in 21)	
21	59.259016	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply	id=0xc010, seq=3/768, ttl=63 (request in 20)	
22	60.261675	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request	id=0xc110, seq=4/1024, ttl=64 (reply in 23)	
▶ Frame 17: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0								0000 00 50 79 66
▶ Ethernet II, Src: 0c:66:71:16:00:02 (0c:66:71:16:00:02), Dst: Private_66:68:04 (00:50:79:66:68:04)								0010 00 54 10 be
▶ Destination: Private_66:68:04 (00:50:79:66:68:04)								0020 01 0a 00 00
▶ Source: 0c:66:71:16:00:02 (0c:66:71:16:00:02)								0030 0e 0f 10 11
Type: IPv4 (0x0800)								0040 1e 1f 20 21
[Stream index: 4]								0050 2e 2f 30 31
▶ Internet Protocol Version 4, Src: 172.16.20.10, Dst: 64.100.1.10								0060 3e 3f
0100 = Version: 4								
.... 0101 = Header Length: 20 bytes (5)								
▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)								
Total Length: 84								
Identification: 0x10be (4286)								
▶ 000. = Flags: 0x0								
...0 0000 0000 0000 = Fragment Offset: 0								
Time to Live: 63								
Protocol: ICMP (1)								
Header Checksum: 0x6963 [validation disabled]								
[Header checksum status: Unverified]								
Source Address: 172.16.20.10								
Destination Address: 64.100.1.10								
[Stream index: 0]								
▶ Internet Control Message Protocol								
Type: 0 (Echo (ping) reply)								
Code: 0								
Checksum: 0x69fa [correct]								
[Checksum Status: Good]								
Identifier (BE): 48656 (0xbe10)								
Identifier (LE): 4286 (0x10be)								
Sequence Number (BE): 1 (0x0001)								
Sequence Number (LE): 256 (0x0100)								
[Request frame: 16]								
[Response time: 3,076 ms]								
▶ Data (56 bytes)								

Рис. 18: ICMP Echo reply и параметры пакета

ICMPv6: Echo request / reply (IPv6)

28	75.328371	2001:db8:c0de:11::a	2001:db8:c0de:13::a	ICMPv6	118 Echo (ping) request id=0xd010, seq=1, hop limit=64 (reply in 29)		
29	75.330878	2001:db8:c0de:13::a	2001:db8:c0de:11::a	ICMPv6	118 Echo (ping) reply id=0xd010, seq=1, hop limit=62 (request in 28)		
30	76.332224	2001:db8:c0de:11::a	2001:db8:c0de:13::a	ICMPv6	118 Echo (ping) request id=0xd010, seq=2, hop limit=64 (reply in 31)		
31	76.333638	2001:db8:c0de:13::a	2001:db8:c0de:11::a	ICMPv6	118 Echo (ping) reply id=0xd010, seq=2, hop limit=62 (request in 30)		
32	77.335133	2001:db8:c0de:11::a	2001:db8:c0de:13::a	ICMPv6	118 Echo (ping) request id=0xd010, seq=3, hop limit=64 (reply in 33)		
33	77.338618	2001:db8:c0de:13::a	2001:db8:c0de:11::a	ICMPv6	118 Echo (ping) reply id=0xd010, seq=3, hop limit=62 (request in 32)		
34	78.340640	2001:db8:c0de:11::a	2001:db8:c0de:13::a	ICMPv6	118 Echo (ping) request id=0xd010, seq=4, hop limit=64 (reply in 35)		
▶ Frame 28: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0							0000 0c
▼ Ethernet II, Src: Private_66:68:04 (00:50:79:66:68:04), Dst: 0c:96:6a:f9:00:02 (0c:96:6a:f9:00:02)							0010 00
▶ Destination: 0c:96:6a:f9:00:02 (0c:96:6a:f9:00:02)							0020 00
▶ Source: Private_66:68:04 (00:50:79:66:68:04)							0030 00
Type: IPv6 (0x86dd)							0040 82
[Stream index: 2]							0050 12
▼ Internet Protocol Version 6, Src: 2001:db8:c0de:11::a, Dst: 2001:db8:c0de:13::a							0060 22
0110 = Version: 6							0070 32
▶ 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)							
.... 0000 0000 0000 0000 = Flow Label: 0x00000							
Payload Length: 64							
Next Header: ICMPv6 (58)							
Hop Limit: 64							
▶ Source Address: 2001:db8:c0de:11::a							
▶ Destination Address: 2001:db8:c0de:13::a							
[Stream index: 4]							
▼ Internet Control Message Protocol v6							
Type: Echo (ping) request (128)							
Code: 0							
Checksum: 0xdaf8 [correct]							
[Checksum Status: Good]							
Identifier: 0xd010							
Sequence: 1							
[Response in: 29]							
▶ Data (66 bytes)							

Рис. 19: ICMPv6 Echo request и поля IPv6/ICMPv6

- **ARP:** соответствие IPv4 ↔ MAC, факт широковещательного запроса, MAC отправителя/цели
- **ICMP:** доступность узла, RTT, идентификатор/sequence, TTL (оценка пути)
- **ICMPv6:** доступность IPv6 узлов, Hop Limit, идентификатор/sequence, адреса источника/назначения

Самостоятельное задание

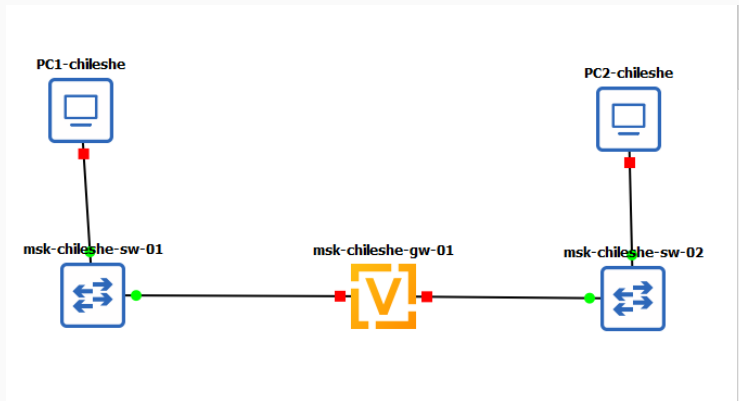


Рис. 20: Топология самостоятельного задания

Таблица адресации (минимальный адрес на интерфейсе маршрутизатора)

Устройство/интерфейс	IPv4/маска	Шлюз (IPv4)	IPv6/префикс
gw-01 eth0	10.10.1.97/27	—	2001:db8:1:1::1/64
gw-01 eth1	10.10.1.17/28	—	2001:db8:1:4::1/64
PC1	10.10.1.100/27	10.10.1.97	2001:db8:1:1::a/64
PC2	10.10.1.20/28	10.10.1.17	2001:db8:1:4::a/64

```
PC1-chileshe - PuTTY
PC1-chileshe> show ip

NAME          : PC1-chileshe[1]
IP/MASK       : 10.10.1.100/27
GATEWAY       : 10.10.1.97
DNS           :
MAC           : 00:50:79:66:68:00
LPORT        : 10008
RHOST:PORT    : 127.0.0.1:10009
MTU           : 1500

PC1-chileshe> show ipv6

NAME          : PC1-chileshe[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 2001:db8:1:1::a/64
DNS             :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:00
LPORT          : 10008
RHOST:PORT     : 127.0.0.1:10009
MTU            : 1500

PC1-chileshe>
```

```
PC2-chileshe> show ip

NAME           : PC2-chileshe[1]
IP/MASK        : 10.10.1.20/28
GATEWAY        : 10.10.1.17
DNS            :
MAC            : 00:50:79:66:68:01
LPORT          : 10010
RHOST:PORT     : 127.0.0.1:10011
MTU            : 1500

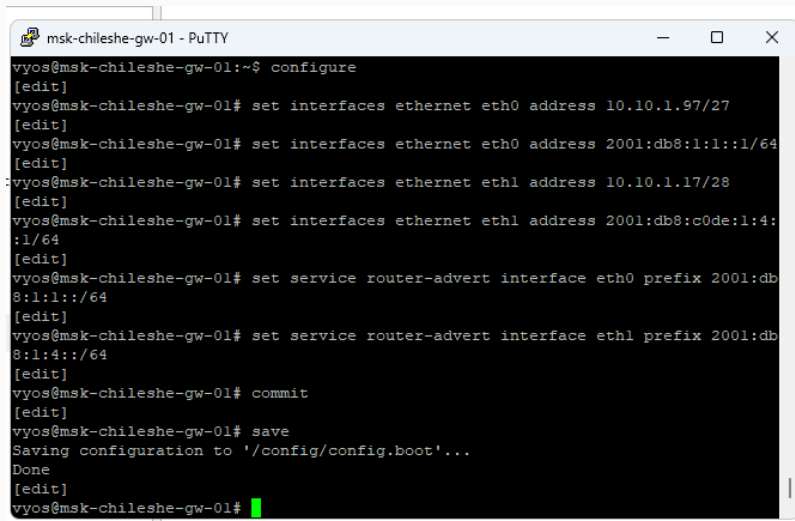
PC2-chileshe> show ipv6

NAME           : PC2-chileshe[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE    : 2001:db8:1:4::a/64
DNS             :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:01
LPORT          : 10010
RHOST:PORT     : 127.0.0.1:10011
MTU            : 1500

PC2-chileshe> 
```

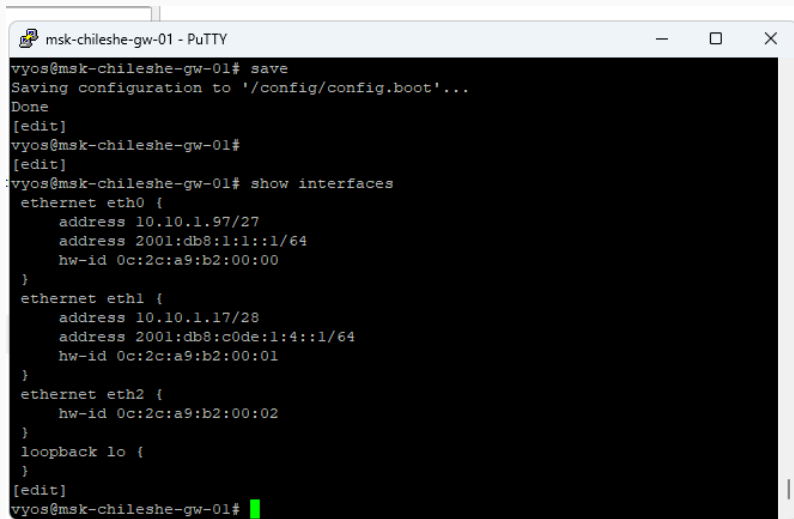
Рис. 22: show ip и show ipv6 на PC2

Настройка адресации на VyOS (gw-01)



```
msh-chileshe-gw-01 - PuTTY
vyos@msh-chileshe-gw-01:~$ configure
[edit]
vyos@msh-chileshe-gw-01# set interfaces ethernet eth0 address 10.10.1.97/27
[edit]
vyos@msh-chileshe-gw-01# set interfaces ethernet eth0 address 2001:db8:1:1::1/64
[edit]
vyos@msh-chileshe-gw-01# set interfaces ethernet eth1 address 10.10.1.17/28
[edit]
vyos@msh-chileshe-gw-01# set interfaces ethernet eth1 address 2001:db8:c0de:1:4:
:1/64
[edit]
vyos@msh-chileshe-gw-01# set service router-advert interface eth0 prefix 2001:db
8:1:1::/64
[edit]
vyos@msh-chileshe-gw-01# set service router-advert interface eth1 prefix 2001:db
8:1:4::/64
[edit]
vyos@msh-chileshe-gw-01# commit
[edit]
vyos@msh-chileshe-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msh-chileshe-gw-01#
```

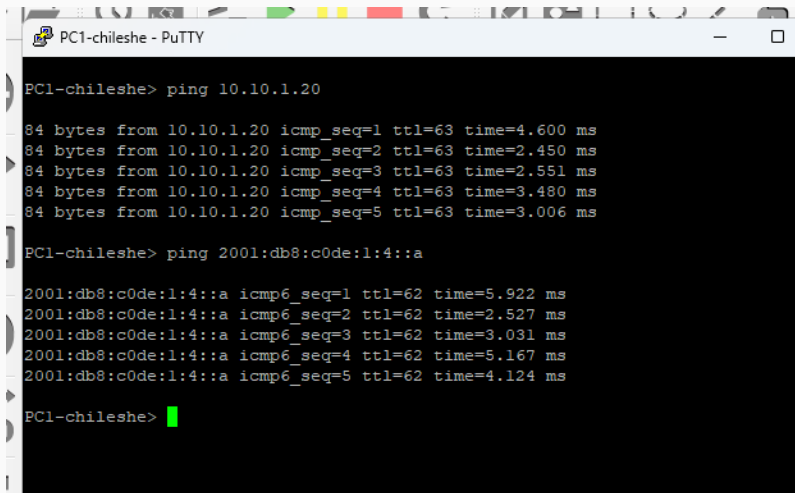
Рис. 23: Настройка IPv4/IPv6 и RA на VyOS



```
msk-chileshe-gw-01 - PuTTY
vyos@msk-chileshe-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-chileshe-gw-01#
[edit]
vyos@msk-chileshe-gw-01# show interfaces
  ethernet eth0 {
    address 10.10.1.97/27
    address 2001:db8:1:1::1/64
    hw-id 0c:2c:a9:b2:00:00
  }
  ethernet eth1 {
    address 10.10.1.17/28
    address 2001:db8:c0de:1:4::1/64
    hw-id 0c:2c:a9:b2:00:01
  }
  ethernet eth2 {
    hw-id 0c:2c:a9:b2:00:02
  }
  loopback lo {
  }
[edit]
vyos@msk-chileshe-gw-01#
```

Рис. 24: show interfaces (адреса на eth0/eth1)

Проверка связности (ping)



```
PC1-chileshe> ping 10.10.1.20

84 bytes from 10.10.1.20 icmp_seq=1 ttl=63 time=4.600 ms
84 bytes from 10.10.1.20 icmp_seq=2 ttl=63 time=2.450 ms
84 bytes from 10.10.1.20 icmp_seq=3 ttl=63 time=2.551 ms
84 bytes from 10.10.1.20 icmp_seq=4 ttl=63 time=3.480 ms
84 bytes from 10.10.1.20 icmp_seq=5 ttl=63 time=3.006 ms

PC1-chileshe> ping 2001:db8:c0de:1:4::a

2001:db8:c0de:1:4::a icmp6_seq=1 ttl=62 time=5.922 ms
2001:db8:c0de:1:4::a icmp6_seq=2 ttl=62 time=2.527 ms
2001:db8:c0de:1:4::a icmp6_seq=3 ttl=62 time=3.031 ms
2001:db8:c0de:1:4::a icmp6_seq=4 ttl=62 time=5.167 ms
2001:db8:c0de:1:4::a icmp6_seq=5 ttl=62 time=4.124 ms

PC1-chileshe> 
```

Рис. 25: Ping по IPv4 и IPv6 между подсетями

Проверка маршрута (trace)

```
PC1-chileshe>
PC1-chileshe> trace 10.10.1.20
trace to 10.10.1.20, 8 hops max, press Ctrl+C to stop
 1  10.10.1.97    3.496 ms  1.578 ms  1.271 ms
 2  *10.10.1.20   3.576 ms (ICMP type:3, code:3, Destination port unreachable)

PC1-chileshe> trace 2001:db8:c0de:1:4::a
trace to 2001:db8:c0de:1:4::a, 64 hops max
 1 2001:db8:1:1::1  3.840 ms  1.782 ms  1.187 ms
 2 2001:db8:c0de:1:4::a  3.817 ms  1.541 ms  1.261 ms

PC1-chileshe> █
```

Рис. 26: Trace по IPv4 и IPv6

Вывод

- Выполнено разбиение адресного пространства **IPv4 (VLSM)** и **IPv6**
- Настроена сеть в GNS3 с **FRR (IPv4)** и **VyOS (IPv6)**
- Подтверждена работа **Dual Stack** и изоляция подсетей IPv4/IPv6
- Проанализирован трафик **ARP, ICMP, ICMPv6** в Wireshark